CSC3100 Data Structures

Assignment 1: LeetCode 118 - Pascal's Triangle

Deadline: 30 June 2026, 23:59 (UTC+8)

Language: Python 3.11 **Total:** 100 marks

Problem Description

Given an integer **numRows**, return the first numRows of Pascal's Triangle. In Pascal's Triangle, each number is the sum of the two numbers directly above it.

Required Interface

```
def generate(numRows: int) -> list[list[int]]
```

Constraints: 1 <= numRows <= 20. Do not use third-party libraries, network access, or local file access. Each test case has a two-second limit.

Public Examples

Input	Expected output
numRows = 1	[[1]]
numRows = 3	[[1], [1, 1], [1, 2, 1]]
numRows = 5	[[1], [1, 1], [1, 2, 1], [1, 3, 3, 1], [1, 4, 6, 4, 1]]

Submission Requirements

Upload one ZIP archive named **StudentID_Name_Assignment1.zip**. The archive must contain one top-level folder with the same name. That folder must contain exactly:

Required file	Purpose
代码实现.txt	Python implementation of generate(numRows)
报告.pdf	Implementation report and analysis

Report Requirements

Explain the algorithm, why it is correct, time and space complexity, implementation decisions, and at least one worked example. Clearly identify external assistance and sources.

Assessment and Policies

Code testing contributes 50 marks and the report contributes 50 marks. The detailed rubric is published separately. Late and malformed submissions are flagged for teacher review. AI-writing risk indicators are advisory only and do not automatically reduce marks.